FUEL

BODY FUEL MONITOR

VERSION 1 · USER GUIDE

FUEL shows you what your body is burning right now, tracks your micronutrient intake from food and supplements, and gives you AI-powered nutrition guidance — all in one place.

CONTENTS

SHADRACH SIMON SUNDAR

01 PROFILE SETUP

Four inputs at the top of the Monitor screen personalise every calculation. Change any value and everything updates instantly.

FIELD	WHAT IT DOES
AGE	Sets max heart rate (220-age), adjusts BMR, and drives micronutrient RDAs.
SEX	M or F. Mifflin-St Jeor uses a different constant for each sex.
KG	Body weight. The biggest driver of calories burned at any activity level.
CM	Height. Taller people have slightly higher resting metabolic rates.

02 THE FOUR TABS

The tab switcher has four tabs. The tagline under FUEL changes to match.

- **MON** — Live fuel gauge, meal context, burn breakdown and engine status.
- **PLAN** — Plan activities and get daily macro targets in grams.
- **ADV** — AI nutrition chat. Describe food for macros and micros, or ask what to eat and when based on your current state and today's plan.
- **MICROS** — Supplement tracker with daily intake from food and supplements combined.

03 HEART RATE ZONES

ZONE	% MAX HR	PRIMARY FUEL	WHAT IS HAPPENING
REST	0-50%	80% Fat	Baseline. Fat dominates.
Z1	50-60%	70% Fat	Very light. Fat oxidation active.
Z2	60-70%	65% Fat	Fat burning peak. GLUT4 opens independently of insulin.
Z3	70-80%	50/50	Crossover zone.
Z4	80-90%	75% Glucose	Threshold. Glycolysis dominant.
Z5	90-100%	90% Glucose	Maximum effort. Anaerobic.

04 THE FUEL GAUGE

FUEL	WHAT IT MEANS
FAT	Calories from stored body fat via beta-oxidation. Dominant at rest and low intensity.
GLUCOSE	Calories from carbohydrates — recent meal or glycogen stores.
PROTEIN	Amino acids via gluconeogenesis. Only after extended fasting or very high intensity.

WHAT THE PERCENTAGE MEANS

If glucose shows **28%** and you burn 80 kcal/hr, **22 of those 80 calories** are coming from glucose right now. A proportion of live calorie burn — not a blood glucose reading.

05 CURRENT ACTIVITY

GROUP	ACTIVITIES	ZONE
REST & DAILY	Sleep · Relax · Desk Work · Standing	Rest
AEROBIC	Slow Walk · Brisk Walk · Zone 2	Z1-Z2
HIGH INTENSITY	Skipping · HIIT · Sprinting · Max Effort	Z4-Z5

06 MEAL CONTEXT

Records two meals to build a complete picture of your current fuel state.

1	**PREVIOUS MEAL — TIME** Set the time you ate your earlier meal. Tap the ■ icon to open the picker.
2	**PREVIOUS MEAL — ACTIVITIES** Log activities using + ADD ACTIVITY. Multiple blocks supported.
3	**PREVIOUS MEAL — WHAT YOU ATE** Enter carbs, fat, protein in grams. Kcal calculated automatically.
4	**LAST MEAL — TIME** The biggest single driver of your current fuel mix. Tap NOW to set to current time instantly.
5	**LAST MEAL — ACTIVITIES** Zone 4-5 before eating depletes glycogen — post-meal glucose stores into muscle. No activity means slower clearance.
6	**LAST MEAL — WHAT YOU ATE** Affects the fuel mix for the first 3-4 hours after eating.

07 ACTIVITY BLOCKS

Log activities with exact FROM and TO times. Duration x zone intensity = glycogen depletion score.

- **+ ADD ACTIVITY** creates a block. Set FROM and TO times using the native iOS wheel.
- **Duration and kcal** appear automatically once both times are set.
- **Remove** by tapping X on the right of the activity dropdown.
- **Multiple blocks** supported. Total glycogen depletion calculated across all blocks.

08 FASTING WINDOW

WINDOW	COLOUR	WHAT IT MEANS
Under 2h	Red	Last meal still processing. New glucose has limited storage.

WINDOW	COLOUR	WHAT IT MEANS
2-4h	Amber	Transitioning. Some glycogen space opening.
4-6h	Blue	Last meal cleared. Body ready to handle glucose efficiently.
6-10h	Green	Running on fat and liver stores.
10h+	Green	Extended fast. Liver making glucose from stored glycogen.

09 MOVE TO PREVIOUS MEAL

1 **FIRST TAP — CONFIRM**
Button turns amber. Resets after 4 seconds if not confirmed.

2 **SECOND TAP — EXECUTES**
Time, macros, and all activity blocks copy into Previous Meal. Last Meal clears.

10 WHAT IS DRIVING YOUR BURN

ROW	WHAT IT SHOWS
NOW	Current activity and base kcal/hr from BMR x MET.
BEFORE MEAL	Activity blocks — total minutes, kcal, glycogen depletion.
EPOC BOOST	Only after high intensity. % above baseline and source activity.
FUEL SHIFT	Net effect of pre-meal activity on fuel mix.
FASTING WINDOW	Only shown in first 4 hours after eating.
PREV MEAL	Previous meal kcal, macro split, influence on current conditions.
NET BURN RATE	Final kcal/hr with exact fuel split.

11 EPOC — THE AFTER-BURN

ACTIVITY	+% HOUR 1	+% HOUR 2	+% HOUR 4	FADES BY
Zone 5	+22%	+16%	+9%	8 hrs
Zone 4	+16%	+11%	+6%	6 hrs
Zone 2	+6%	+3%	+1%	4 hrs
Rest	None	None	None	—

12 ENGINE STATUS

Pure body description. Updates every minute. Four structured sentences: fuel state, time context, pre-meal activity effect, fasting window.

13 ACTION CARD

SITUATION	ACTION SHOWN
Post-HIIT within 2h, now resting	Session completed. Eat 25-30g protein now.
Zone 4/5 currently active	Session active — eat 25-30g protein within 2 hours.
Zone 2 currently active	Protein within 60 minutes after this session.
Under 2h post-meal, glucose > 45%	A 10-minute walk now activates muscle glucose uptake.
2-4h post-meal	Let this meal finish processing before eating again.
4-6h post-meal	Good window to eat if hungry — fat or protein first.
6h+ post-meal	Running on stored fat — no need to eat unless genuinely hungry.

14 BODY FUEL PLANNER

Plan activities for the day and get a personalised macro target in grams. Duration inputs support min and hr.

1 SET YOUR GOAL

Maintain — eat roughly what you burn. **Lose Fat** — 500 kcal deficit. **Build** — 250 kcal surplus.

2 ADD ACTIVITIES

Tap + ADD ACTIVITY. Select activity and duration. Use the min/hr selector to switch units.

3 READ THE BURN

Base metabolic burn for sedentary hours plus each activity block.

4 GET MACRO TARGETS

Carbs, fat, and protein in grams. Split adjusts by activity type.

5 TIMING GUIDE

When to eat each macro relative to planned activities.

PLANNER FEEDS THE ADVISOR

Activities planned here are included in the Advisor's context. If you plan a Zone 4 session at 18:00, the Advisor accounts for this when answering what to eat and when — it will time carbs around your planned session.

15 ADVISOR TAB (ADV)

AI-powered nutrition chat using Google Gemini. The Advisor knows your full current state — profile, activity, meal timing, fasting window, and today's planned activities.

- **Describe what you ate** — Gemini calculates macros per item, checks all 8 tracked micros, and reports every one that is present in the meal.
- **MACRO BREAKDOWN card** — CARBS / FAT / PROTEIN / kcal. BREAKDOWN button expands per-item macros.
- **MICRO CONTRIBUTIONS card** — shows every micro present with amount in Fuel units and percentage of your daily target. All values added to MICROS tab automatically.
- → **Apply buttons** — PREVIOUS MEAL or LAST MEAL. Applies macros and switches to Monitor.
- **What should I eat and when?** chip — responds with: RIGHT NOW: specific food NEXT MEAL: clock time and food TRAINING: timing advice based on your planned activities

RETRY — appears on error. Resends your last message when Google servers are busy.

ALL 8 MICROS CHECKED

For every food query, the Advisor explicitly checks all 8 tracked micros: Vitamin D3, Magnesium, Zinc, B12, Iron, Potassium, Omega-3, and K2. Each one is evaluated individually against the foods in your meal.

ACCURACY

Macros accurate to ±15% for well-known foods. Micro values are directionally useful for daily tracking — not clinically precise. Values vary by food source, preparation method, and growing conditions.

16 MICROS TAB

Tracks eight key micronutrients. Each card shows daily intake from food logged in the Advisor and supplements manually recorded.

UNIT CONVERSION

MICRO	GEMINI REPORTS	FUEL STORES	CONVERSION
Vitamin D3	mcg (FDA standard)	IU	x 40 (1 mcg = 40 IU)
Magnesium	mg	mg	None
Zinc	mg	mg	None
Vitamin B12	mcg	mcg	None
Iron	mg	mg	None
Potassium	mg or g	mg	x 1000 if g
Omega-3	mg or g	mg	x 1000 if g
Vitamin K2	mcg	mcg	None

EACH CARD SHOWS

- **FROM FOOD (teal bar)** — Micro intake from meals described in the Advisor today. Resets at midnight.
- **SUPPLEMENT (colour bar)** — Active dose based on last recorded supplement x current half-life percentage. Decays in real time.
- ∑ **TOTAL TODAY** — Food + supplement combined vs RDA. Goes green when daily target is met.
- **LOG SUPPLEMENT** — Date and time pickers. Applies on dismiss.
- **HALF-LIFE BAR** — Percentage of supplement dose still active. Decays between app opens.
- **NEXT DOSE** — Exact date/time from last taken plus dosing interval.

SUPPLEMENT	HALF-LIFE	DOSING	RDA
Vitamin D3	24h	Daily	600-800 IU. Many experts recommend 2000-4000 IU with deficiency.
Magnesium	24h	Daily	420mg (M) / 320mg (F). 300+ enzyme reactions.
Zinc	24h	Daily	11mg (M) / 8mg (F). Depleted by stress.
Vitamin B12	6h	Daily	2.4 mcg. Supplements 500-1000mcg due to poor absorption.
Iron	6h	Every 48h	8mg (M/post-menopausal F) / 18mg (F under 51).
Potassium	4h	Daily	3400mg (M) / 2600mg (F). Lost through sweat.
Omega-3	48h	Daily	500mg EPA+DHA.
Vitamin K2	72h	Daily	100 mcg. MK-7 form has 72h half-life.

RESET TODAY

RESET TODAY button clears daily food totals. Auto-resets at midnight.

17 RESEARCH PAPERS & REFERENCES

The calculations in Fuel are based on the following peer-reviewed research:

BASAL METABOLIC RATE

The Mifflin-St Jeor equation used to calculate hourly BMR. Validated as most accurate for healthy adults.

Mifflin MD, St Jeor ST, et al. (1990). A new predictive equation for resting energy expenditure. American Journal of Clinical Nutrition, 51(2), 241-247.

MET VALUES

MET values for all activities. Each kcal/hr is BMR/24 x MET.

Ainsworth BE, Haskell WL, et al. (2011). 2011 Compendium of Physical Activities. Medicine & Science in Sports & Exercise, 43(8), 1575-1581.

MAXIMUM HEART RATE

The 220-minus-age formula for maximum heart rate estimation.

Fox SM, Naughton JP, Haskell WL. (1971). Physical activity and the prevention of coronary heart disease. Annals of Clinical Research, 3(6), 404-432.

EPOC

EPOC magnitude and duration by zone. Zone 5 elevates metabolism by up to 22% for the first hour.

Borsheim E, Bahr R. (2003). Effect of exercise intensity on post-exercise oxygen consumption. Sports Medicine, 33(14), 1037-1060.

ZONE 2 FAT OXIDATION

Peak fat oxidation at Zone 2 intensity.

Venables MC, Achten J, Jeukendrup AE. (2005). Determinants of fat oxidation during exercise. Journal of Applied Physiology, 98(1), 160-167.

GLUT4 AND ZONE 2

Exercise activates GLUT4 independently of insulin.

Holloszy JO. (2003). Forty-year memoir on glucose transport into muscle. American Journal of Physiology, 284(3), E453-E467.

PROTEIN SYNTHESIS TIMING

Protein within 2 hours post high intensity maximally stimulates muscle protein synthesis.

Cribb PJ, Hayes A. (2006). Effects of supplement timing on skeletal muscle hypertrophy. Medicine & Science in Sports & Exercise, 38(11), 1918-1925.

GLYCOGEN RESYNTHESIS

Post-exercise glycogen resynthesis fastest in first 2 hours.

Ivy JL. (1998). Glycogen resynthesis after exercise. International Journal of Sports Medicine, 19(S2), S142-S145.

FASTING METABOLISM

Liver maintains blood glucose from glycogen for 12-16 hours at rest.

Cahill GF. (2006). Fuel metabolism in starvation. Annual Review of Nutrition, 26, 1-22.

VITAMIN D UNIT CONVERSION

FDA mandates Vitamin D declared in mcg. 1 mcg = 40 IU. Basis for Fuel automatic conversion.

FDA Guidance (2016). Nutrition Facts Label Update. 21 CFR Part 101.

VITAMIN D SUPPLEMENTATION

Basis for Vitamin D RDA values and deficiency thresholds.

Holick MF, et al. (2011). Evaluation, treatment, and prevention of vitamin D deficiency. Journal of Clinical Endocrinology and Metabolism, 96(7), 1911-1930.

MAGNESIUM

300+ enzyme reactions. Basis for 420mg/day (M) and 320mg/day (F) RDAs.

Volpe SL. (2013). Magnesium in disease prevention. Advances in Nutrition, 4(3), 378S-383S.

OMEGA-3

EPA and DHA improve insulin receptor sensitivity. Basis for 500mg daily target.

Calder PC. (2013). Omega-3 fatty acids and inflammatory processes. British Journal of Clinical Pharmacology, 75(3), 645-662.

VITAMIN K2 HALF-LIFE

MK-7 half-life approximately 72 hours.

Schurgers LJ, et al. (2007). Vitamin K-containing dietary supplements. Blood, 109(8), 3279-3283.

IRON ABSORPTION

Iron absorption enhanced by Vitamin C, inhibited by calcium.

Hallberg L, et al. (1987). Iron absorption from the whole diet. American Journal of Clinical Nutrition, 46(5), 730-738.